

The Social Equation of Lifespan: Poverty, Health, and Inequality

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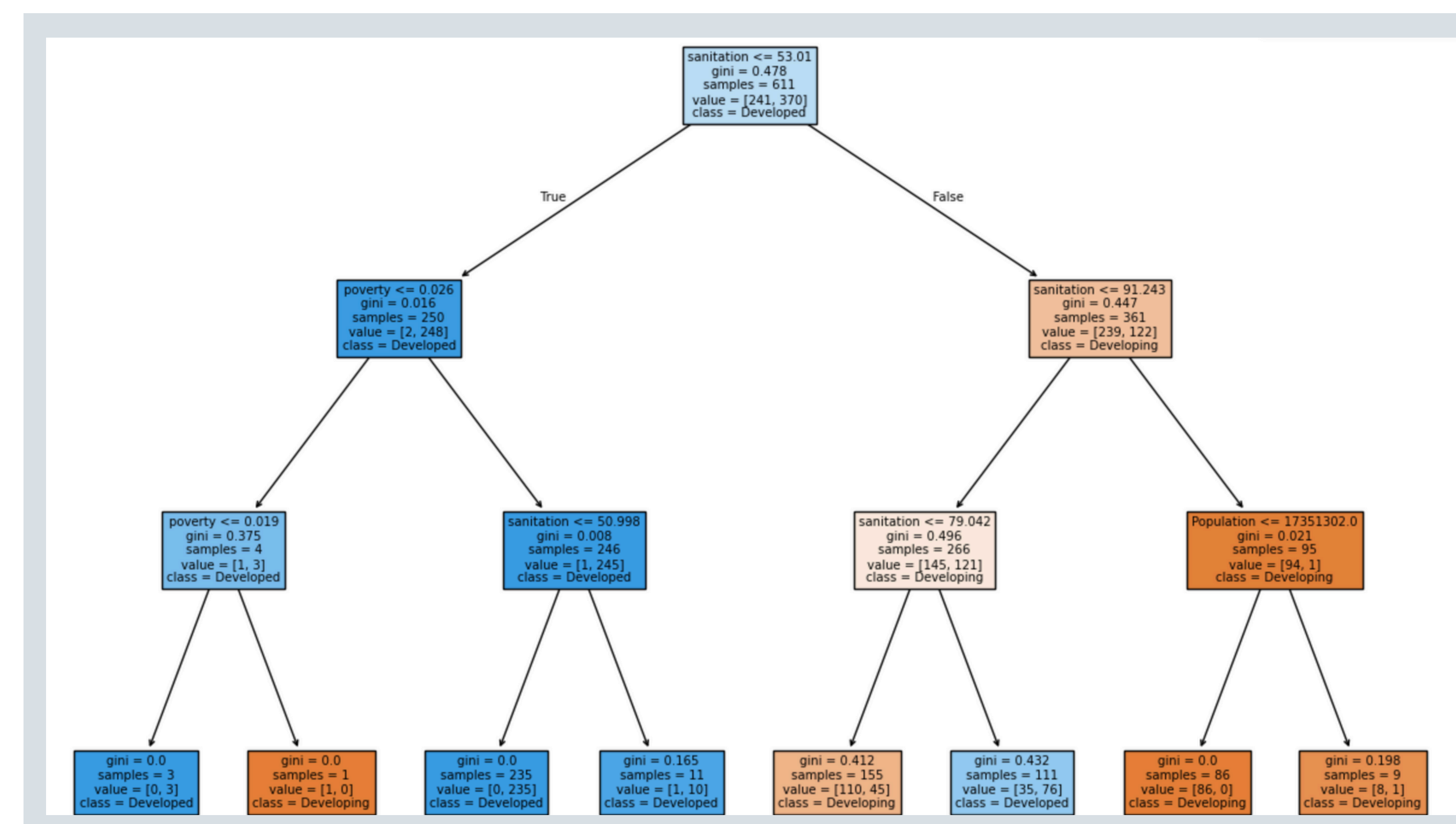
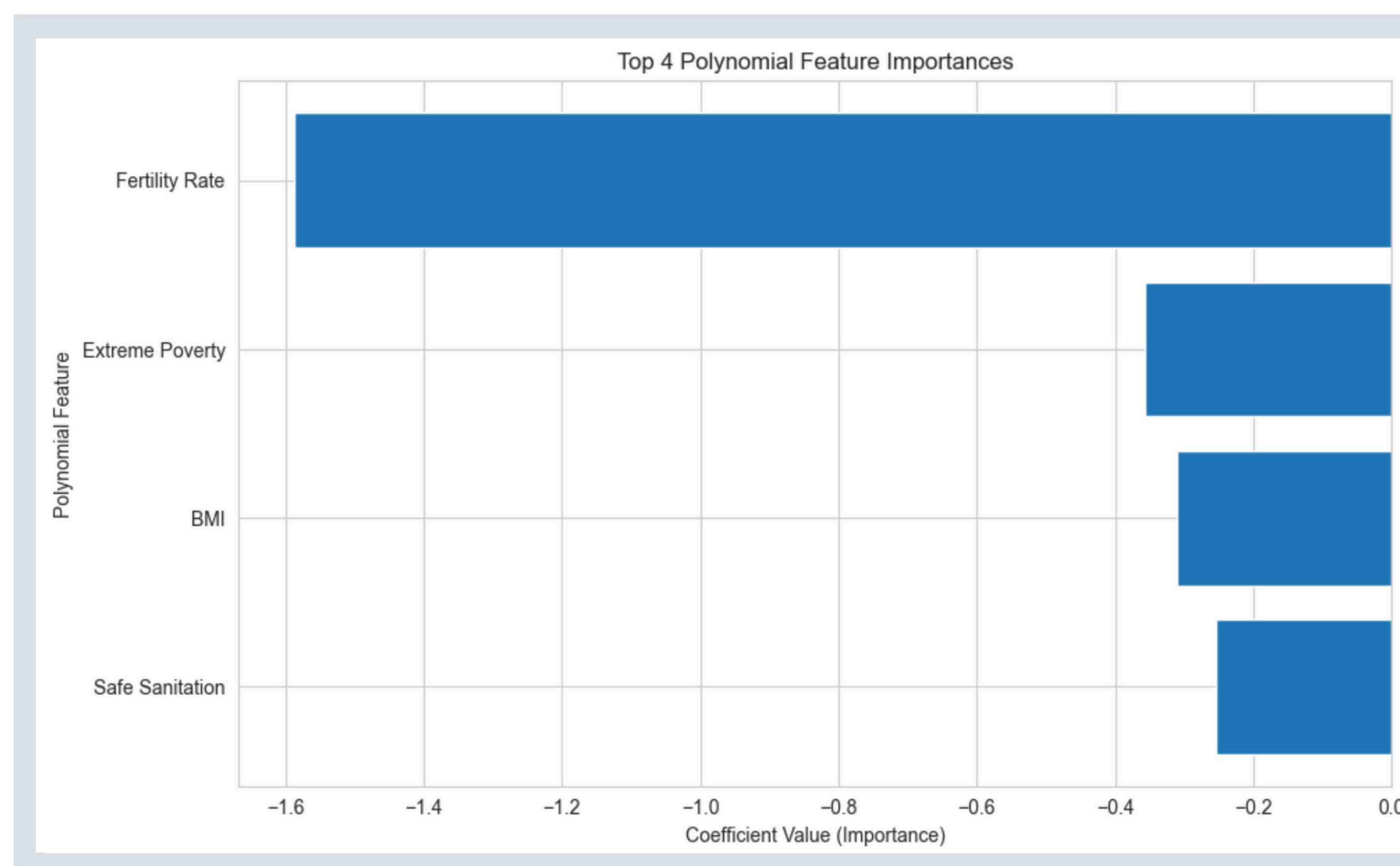
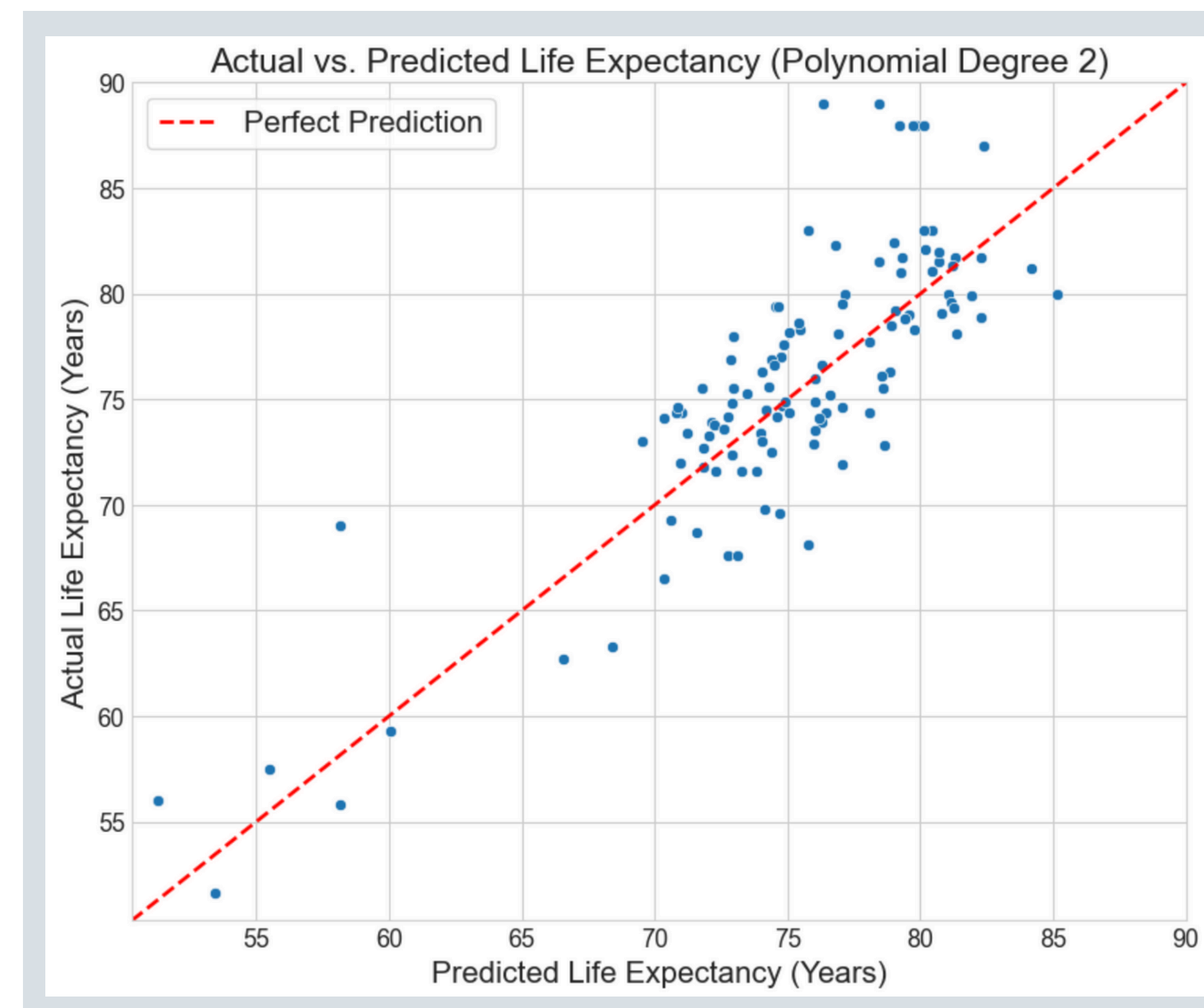
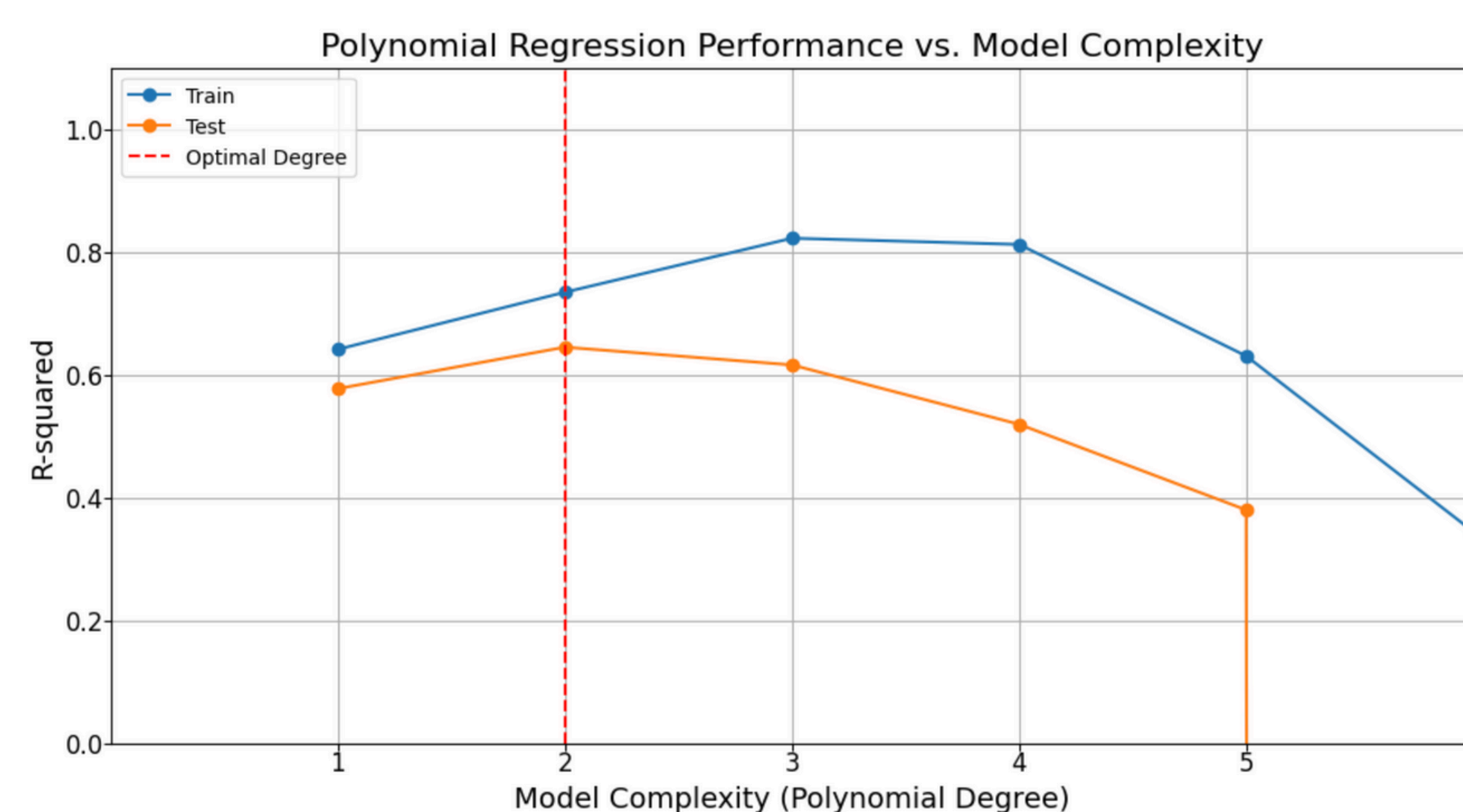
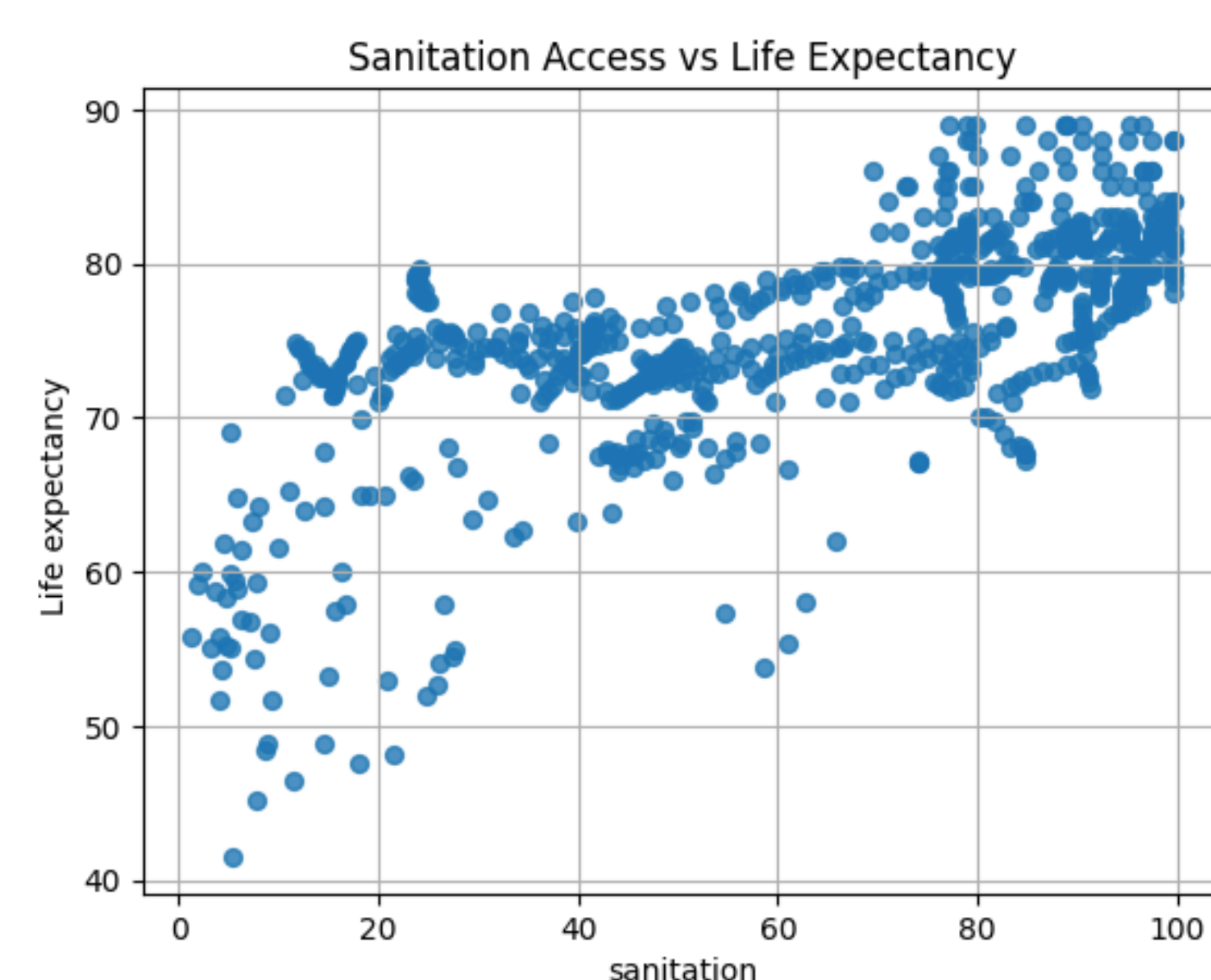
Intro

How do socioeconomic factors, (extreme poverty), income inequality, environmental health indicators, access to safe sanitation, and biological factors, affect life expectancy in various global regions?

By exploring these connections, particularly the roles of fertility rate and sanitation access, this research hopes that it may create evidence-based policies and interventions that can improve public health and prolong life expectancy around the world.

Methodology

- World Health Organization & Our World in Data
- Life Expectancy, Poverty, Economic Inequality, Fertility Rate, Sanitation, BMI



Results

- Polynomial features improved model accuracy over a linear model
- **RMSE** - 3.63 and **R²** - 0.69
- Fertility rate and extreme poverty show the **strongest negative associations** with life expectancy
- Access to safe sanitation and BMI also negatively affect life expectancy, though to a **lesser extent**

Conclusion

- Higher fertility rates, extreme poverty, poor sanitation, and high BMI are key negative predictors of life expectancy, highlighting the critical role of socioeconomic and environmental conditions.
- Our model shows the gaps between different variables and countries, but adding more variables such as climate data, educational data, and additional health data could increase the accuracy of our model.

References

"Life Expectancy (WHO)." *Kaggle*, 2018, www.kaggle.com/datasets/kumarajarshi/life-expectancy-who

Our World in Data. "Our World in Data." *Our World in Data*, 2024, www.ourworldindata.org/.